Importing Libraries

In [ ]:

```
## Importing Libraries ##
import pandas as pd

train_df = pd.read_csv('loan-prediction.csv')
train_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Loan_ID            614 non-null    object 
 1   Gender             601 non-null    object 
 2   Married            611 non-null    object 
 3   Dependents         599 non-null    object 
 4   Education          614 non-null    object 
 5   Self_Employed      582 non-null    object 
 6   ApplicantIncome    614 non-null    int64  
 7   CoapplicantIncome  614 non-null    float64
 8   LoanAmount         592 non-null    float64
 9   Loan_Amount_Term   600 non-null    float64
 10  Credit_History     564 non-null    float64
 11  Property_Area      614 non-null    object 
 12  Loan_Status        614 non-null    object 
dtypes: float64(4), int64(1), object(8)
memory usage: 62.5+ KB
```

In [ ]:

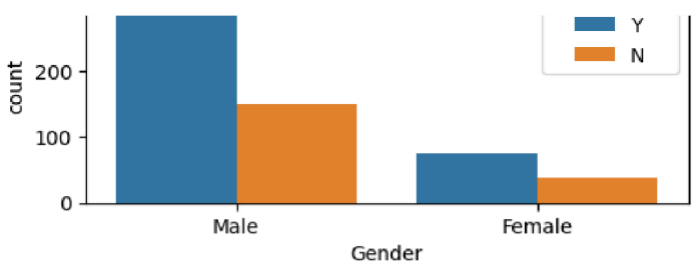
```
############ Count number of Categorical and Numerical Columns ####################
train_df = train_df.drop(columns=['Loan_ID']) ## Dropping Loan ID
categorical_columns = ['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed',
'Property_Area','Credit_History','Loan_Amount_Term']
#categorical_columns = ['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed',
'Property_Area','Loan_Amount_Term']

print(categorical_columns)
numerical_columns = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
print(numerical_columns)
```

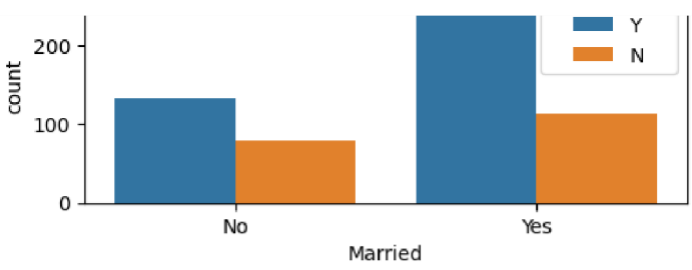
```
['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed', 'Property_Area', 'Credi
t_History', 'Loan_Amount_Term']
['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
```

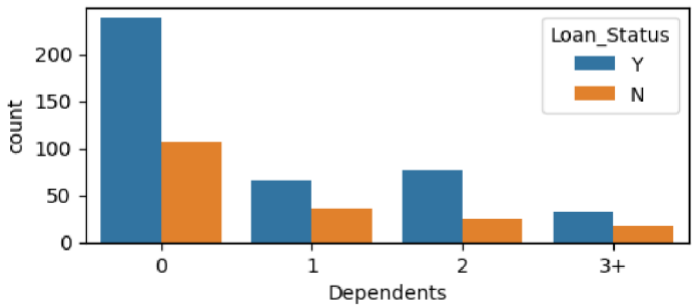
Analyze values assigned to columns

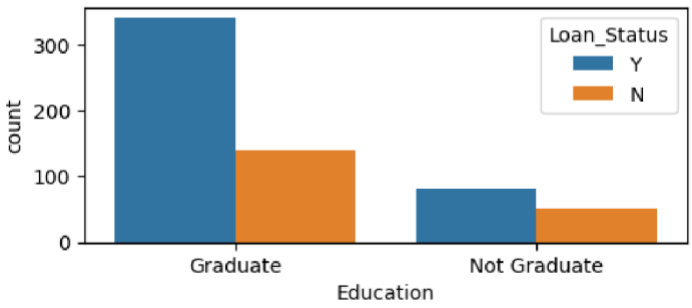
In [ ]:

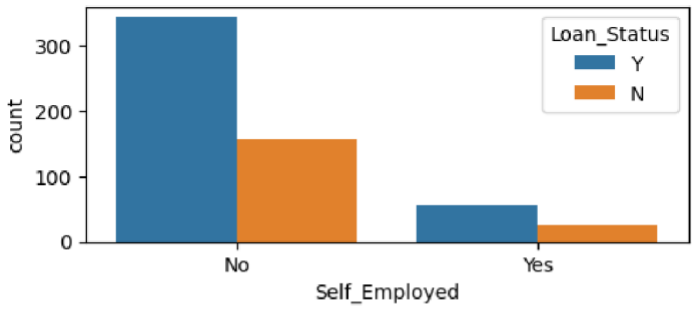
```
### Data Visualization libraries
import seaborn as sns
import matplotlib.pyplot as plt


fig,axes = plt.subplots(4,2,figsize=(12,15))
for idx,cat_col in enumerate(categorical_columns):
    row,col = idx//2,idx%2
    sns.countplot(x=cat_col,data=train_df,hue='Loan_Status',ax=axes[row,col])


plt.subplots_adjust(hspace=1)
```

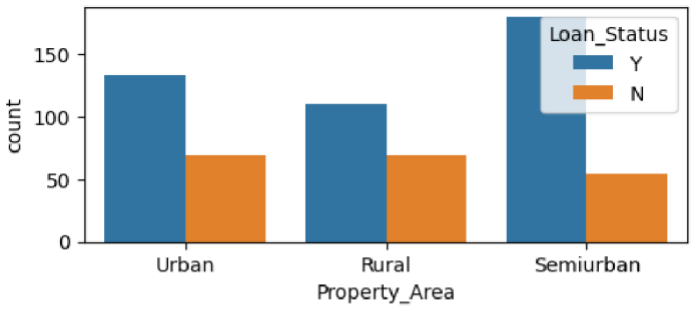


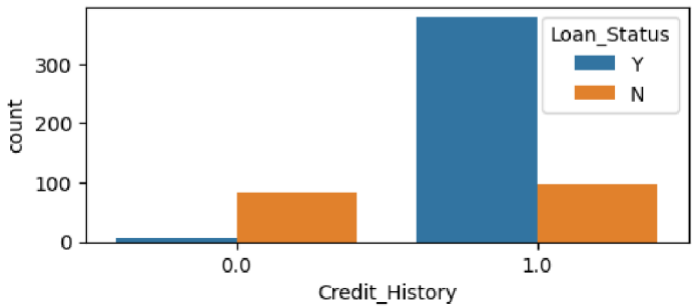


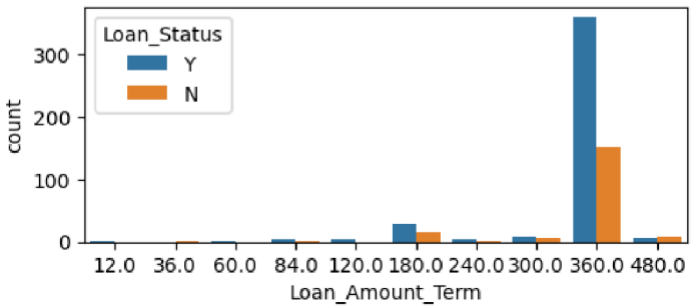










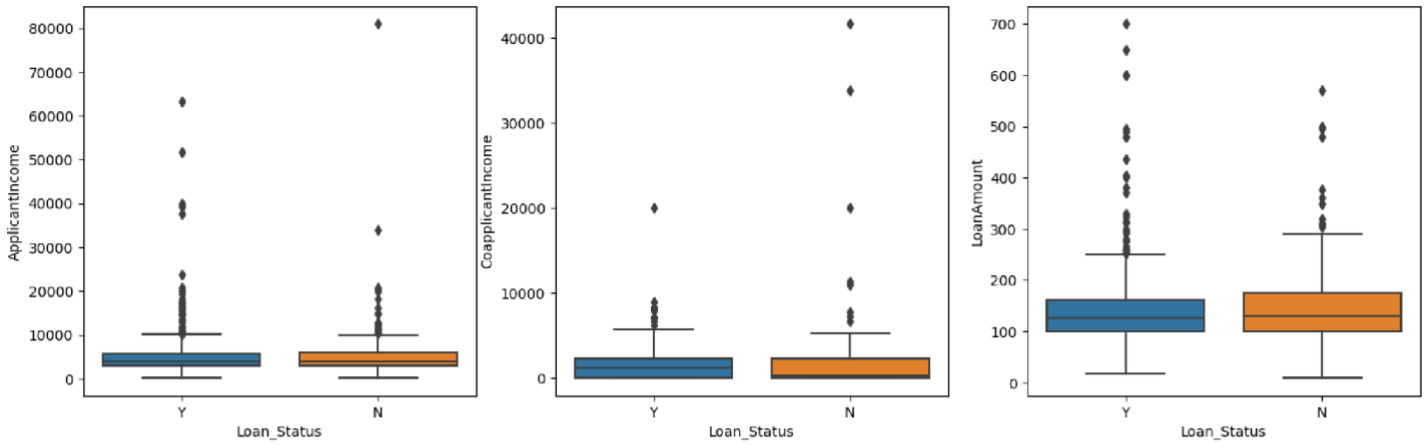


In [ ]:

```
fig,axes = plt.subplots(1,3,figsize=(17,5))
for idx,cat_col in enumerate(numerical_columns):
    sns.boxplot(y=cat_col,data=train_df,x='Loan_Status',ax=axes[idx])

print(train_df[numerical_columns].describe())
plt.subplots_adjust(hspace=1)
```

```
       ApplicantIncome  CoapplicantIncome  LoanAmount
count       614.000000         614.000000  592.000000
mean       5403.459283        1621.245798  146.412162
std        6109.041673        2926.248369   85.587325
min         150.000000           0.000000    9.000000
25%        2877.500000           0.000000  100.000000
50%        3812.500000        1188.500000  128.000000
75%        5795.000000        2297.250000  168.000000
```

```
max        81000.000000       41667.000000   700.000000
```



Preprocessing Data:

Input data needs to be pre-processed before we feed it to model. Following things need to be taken care:

1. **Encoding Categorical Features.**
2. **Imputing missing values**

In [ ]:

```
#### Encoding categrical Features: ##########
train_df_encoded = pd.get_dummies(train_df,drop_first=True)
train_df_encoded.head()
```

Out[ ]:

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Gender_Male	Married_Yes	Depend
0	5849	0.0	NaN	360.0	1.0	True	False	
1	4583	1508.0	128.0	360.0	1.0	True	True	
2	3000	0.0	66.0	360.0	1.0	True	True	
3	2583	2358.0	120.0	360.0	1.0	True	True	
4	6000	0.0	141.0	360.0	1.0	True	False	

In [ ]:

```
########## Split Features and Target Varible ############
X = train_df_encoded.drop(columns='Loan_Status_Y')
y = train_df_encoded['Loan_Status_Y']

################ Splitting into Train -Test Data #######
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,stratify =y,random_st
ate =42)
############### Handling/Imputing Missing values ############
from sklearn.impute import SimpleImputer
imp = SimpleImputer(strategy='mean')
imp_train = imp.fit(X_train)
X_train = imp_train.transform(X_train)
X_test_imp = imp_train.transform(X_test)
```

Model 1: Decision Tree Classifier

In [ ]:

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.model_selection import cross_val_score
from sklearn.metrics import accuracy_score,f1_score


tree_clf = DecisionTreeClassifier()
tree_clf.fit(X_train,y_train)
y_pred = tree_clf.predict(X_train)
print("Training Data Set Accuracy: ", accuracy_score(y_train,y_pred))
print("Training Data F1 Score ", f1_score(y_train,y_pred))

print("Validation Mean F1 Score: ",cross_val_score(tree_clf,X_train,y_train,cv=5,scoring
='f1_macro').mean())
print("Validation Mean Accuracy: ",cross_val_score(tree_clf,X_train,y_train,cv=5,scoring
='accuracy').mean())
```

```
Training Data Set Accuracy:  1.0
Training Data F1 Score  1.0
Validation Mean F1 Score:  0.6590069441404657
Validation Mean Accuracy:  0.7067408781694496
```

Overfitting Problem

We can see from above metrics that Training Accuracy > Test Accuracy with default settings of Decision Tree classifier. Hence, model is overfit. We will try some Hyper-parameter tuning and see if it helps.

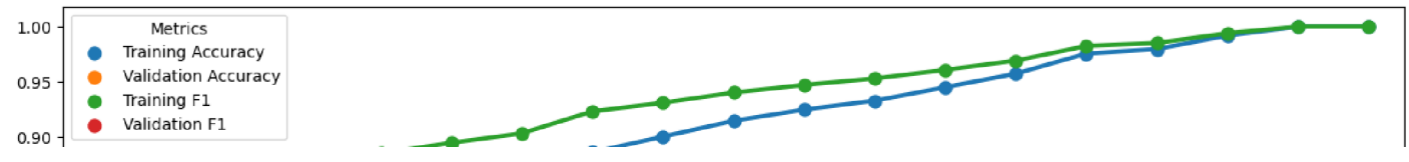
First try tuning 'Max_Depth' of tree

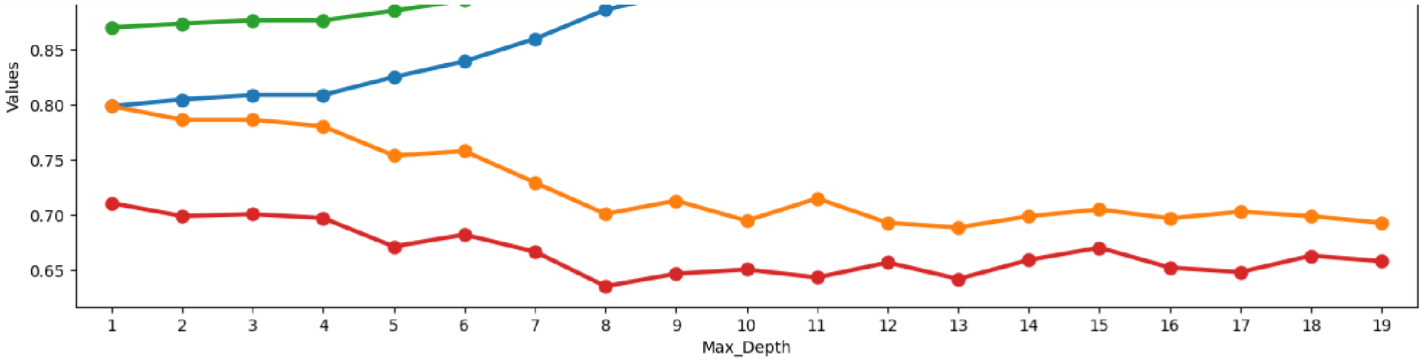
In [ ]:

```
training_accuracy = []
val_accuracy = []
training_f1 = []
val_f1 = []
tree_depths = []

for depth in range(1,20):
    tree_clf = DecisionTreeClassifier(max_depth=depth)
    tree_clf.fit(X_train,y_train)
    y_training_pred = tree_clf.predict(X_train)

    training_acc = accuracy_score(y_train,y_training_pred)
    train_f1 = f1_score(y_train,y_training_pred)
    val_mean_f1 = cross_val_score(tree_clf,X_train,y_train,cv=5,scoring='f1_macro').mean
()
    val_mean_accuracy = cross_val_score(tree_clf,X_train,y_train,cv=5,scoring='accuracy'
).mean()

    training_accuracy.append(training_acc)
    val_accuracy.append(val_mean_accuracy)
    training_f1.append(train_f1)
    val_f1.append(val_mean_f1)
    tree_depths.append(depth)


Tuning_Max_depth = {"Training Accuracy": training_accuracy, "Validation Accuracy": val_a
ccuracy, "Training F1": training_f1, "Validation F1":val_f1, "Max_Depth": tree_depths }
Tuning_Max_depth_df = pd.DataFrame.from_dict(Tuning_Max_depth)

plot_df = Tuning_Max_depth_df.melt('Max_Depth',var_name='Metrics',value_name="Values")
fig,ax = plt.subplots(figsize=(15,5))
sns.pointplot(x="Max_Depth", y="Values",hue="Metrics", data=plot_df,ax=ax)
```

Out[ ]:

```
<Axes: xlabel='Max_Depth', ylabel='Values'>
```





From above graph, we can conclude that keeping 'Max_Depth' = 3 will yield optimum Test accuracy and F1 score Optimum Test Accuracy ~ 0.805; Optimum F1 Score: ~0.7

Visualizing Decision Tree with Max Depth = 3

In [ ]:

```
import graphviz
from sklearn import tree

tree_clf = tree.DecisionTreeClassifier(max_depth = 3)
tree_clf.fit(X_train,y_train)
dot_data = tree.export_graphviz(tree_clf,feature_names = X.columns.tolist())
graph = graphviz.Source(dot_data)
graph
```

In [ ]:

```
pip install graphviz
```

```
Requirement already satisfied: graphviz in c:\users\user\anaconda3\lib\site-packages (0.2
0.3)
Note: you may need to restart the kernel to use updated packages.
```

From above tree, we could see that some of the leafs have less than 5 samples hence our classifier might overfit. We can sweep hyper-parameter 'min_samples_leaf' to further improve test accuracy by keeping max_depth to 3

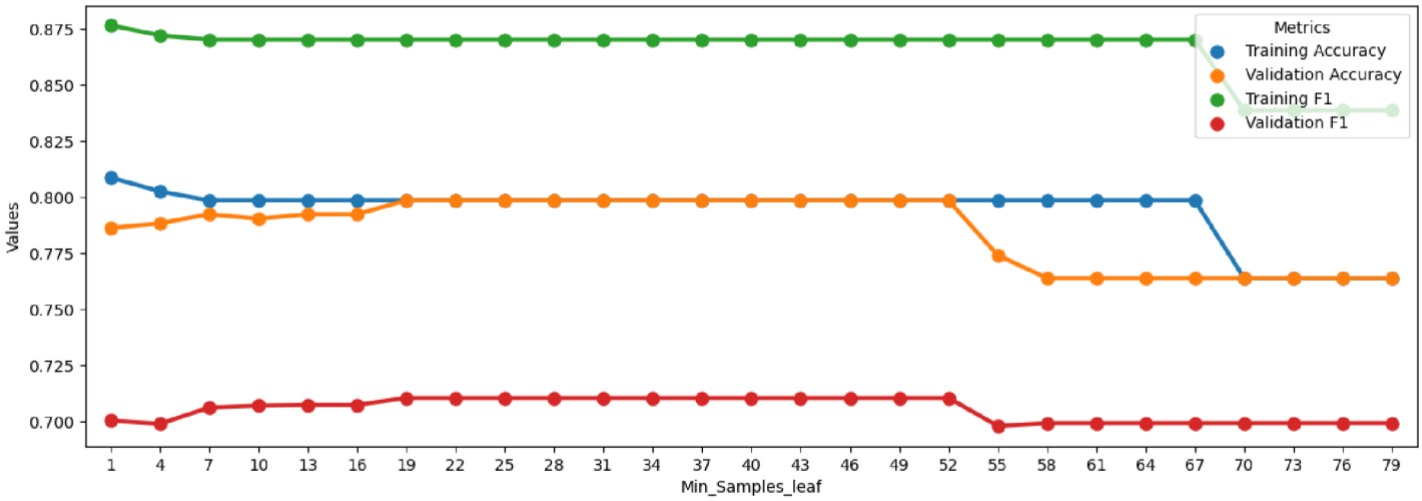
In [ ]:

```
training_accuracy = []
val_accuracy = []
training_f1 = []
val_f1 = []
min_samples_leaf = []
import numpy as np
for samples_leaf in range(1,80,3): ### Sweeping from 1% samples to 10% samples per leaf
    tree_clf = DecisionTreeClassifier(max_depth=3,min_samples_leaf = samples_leaf)
    tree_clf.fit(X_train,y_train)
    y_training_pred = tree_clf.predict(X_train)

    training_acc = accuracy_score(y_train,y_training_pred)
    train_f1 = f1_score(y_train,y_training_pred)
    val_mean_f1 = cross_val_score(tree_clf,X_train,y_train,cv=5,scoring='f1_macro').mean
()
    val_mean_accuracy = cross_val_score(tree_clf,X_train,y_train,cv=5,scoring='accuracy'
).mean()

    training_accuracy.append(training_acc)
    val_accuracy.append(val_mean_accuracy)
    training_f1.append(train_f1)
    val_f1.append(val_mean_f1)
    min_samples_leaf.append(samples_leaf)


Tuning_min_samples_leaf = {"Training Accuracy": training_accuracy, "Validation Accuracy":
val_accuracy, "Training F1": training_f1, "Validation F1":val_f1, "Min_Samples_leaf": min
```

```
_samples_leaf }
Tuning_min_samples_leaf_df = pd.DataFrame.from_dict(Tuning_min_samples_leaf)

plot_df = Tuning_min_samples_leaf_df.melt('Min_Samples_leaf',var_name='Metrics',value_nam
e="Values")
fig,ax = plt.subplots(figsize=(15,5))
sns.pointplot(x="Min_Samples_leaf", y="Values",hue="Metrics", data=plot_df,ax=ax)
```

Out[ ]:

```
<Axes: xlabel='Min_Samples_leaf', ylabel='Values'>
```



From above plot, we will choose Min_Samples_leaf to 35 to improve test accuracy.

Let's use this Decision Tree classifier on unseen test data and evaluate Test Accuracy, F1 Score and Confusion Matrix

In [ ]:

```
from sklearn.metrics import confusion_matrix
tree_clf = DecisionTreeClassifier(max_depth=3,min_samples_leaf = 35)
tree_clf.fit(X_train,y_train)
y_pred = tree_clf.predict(X_test_imp)
print("Test Accuracy: ",accuracy_score(y_test,y_pred))
print("Test F1 Score: ",f1_score(y_test,y_pred))
print("Confusion Matrix on Test Data")
pd.crosstab(y_test, y_pred, rownames=['True'], colnames=['Predicted'], margins=True)
```

```
Test Accuracy:  0.8536585365853658
Test F1 Score:  0.903225806451613
Confusion Matrix on Test Data
```

Out[ ]:

Predicted	False	True	All
True			
False	21	17	38
True	1	84	85
All	22	101	123

Mis-classifications

It can be seen that majority of the misclassifications are happening because of Loan Reject applicants being classified as Accept.

Model 2: Random Forest Classifier

In [ ]:

```
from sklearn.ensemble import RandomForestClassifier

rf_clf = RandomForestClassifier(n_estimators=100,max_depth=3,min_samples_leaf = 10)
rf_clf.fit(X_train,y_train)
y_pred = rf_clf.predict(X_train)
print("Train F1 Score ", f1_score(y_train,y_pred))
print("Train Accuracy ", accuracy_score(y_train,y_pred))

print("Validation Mean F1 Score: ",cross_val_score(rf_clf,X_train,y_train,cv=5,scoring='
f1_macro').mean())
print("Validation Mean Accuracy: ",cross_val_score(rf_clf,X_train,y_train,cv=5,scoring='
accuracy').mean())
```

```
Train F1 Score  0.8699080157687253
Train Accuracy  0.7983706720977597
Validation Mean F1 Score:  0.7105036634489533
Validation Mean Accuracy:  0.79428983714698
```

Random Forest: Test Data Evaluation

In [ ]:

```
y_pred = rf_clf.predict(X_test_imp)
print("Test Accuracy: ",accuracy_score(y_test,y_pred))
print("Test F1 Score: ",f1_score(y_test,y_pred))
print("Confusion Matrix on Test Data")
pd.crosstab(y_test, y_pred, rownames=['True'], colnames=['Predicted'], margins=True)
```

```
Test Accuracy:  0.8536585365853658
Test F1 Score:  0.903225806451613
Confusion Matrix on Test Data
```

Out[ ]:

Predicted	False	True	All
True			
False	21	17	38
True	1	84	85
All	22	101	123

Random Forest gives same results as Decision Tree Classifier. Finally, we will try Logistic Regression Model by sweeping threshold values.

Model 3: Logistic Regression

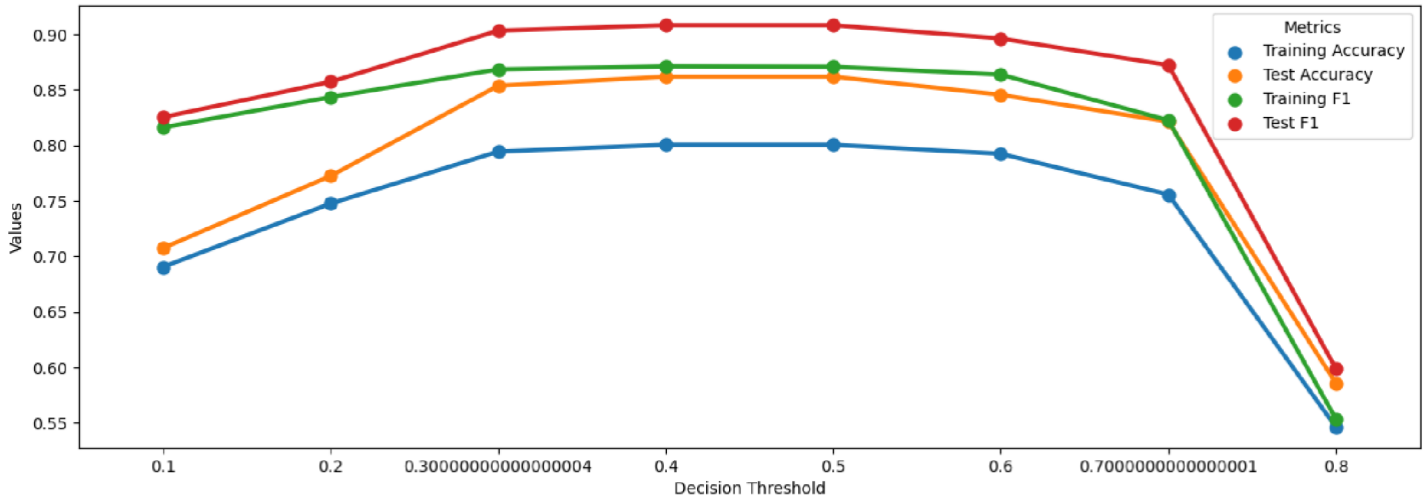
In [ ]:

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.model_selection import cross_val_predict

train_accuracies = []
train_f1_scores = []
test_accuracies = []
test_f1_scores = []
thresholds = []

#for thresh in np.linspace(0.1,0.9,8): ## Sweeping from threshold of 0.1 to 0.9
for thresh in np.arange(0.1,0.9,0.1): ## Sweeping from threshold of 0.1 to 0.9
    logreg_clf = LogisticRegression(solver='liblinear')
    logreg_clf.fit(X_train,y_train)

    y_pred_train_thresh = logreg_clf.predict_proba(X_train)[:,1]
    y_pred_train = (y_pred_train_thresh > thresh).astype(int)
```

```
    train_acc = accuracy_score(y_train,y_pred_train)
    train_f1 = f1_score(y_train,y_pred_train)

    y_pred_test_thresh = logreg_clf.predict_proba(X_test_imp)[:,1]
    y_pred_test = (y_pred_test_thresh > thresh).astype(int)

    test_acc = accuracy_score(y_test,y_pred_test)
    test_f1 = f1_score(y_test,y_pred_test)

    train_accuracies.append(train_acc)
    train_f1_scores.append(train_f1)
    test_accuracies.append(test_acc)
    test_f1_scores.append(test_f1)
    thresholds.append(thresh)


Threshold_logreg = {"Training Accuracy": train_accuracies, "Test Accuracy": test_accuraci
es, "Training F1": train_f1_scores, "Test F1":test_f1_scores, "Decision Threshold": thres
holds }
Threshold_logreg_df = pd.DataFrame.from_dict(Threshold_logreg)

plot_df = Threshold_logreg_df.melt('Decision Threshold',var_name='Metrics',value_name="Va
lues")
fig,ax = plt.subplots(figsize=(15,5))
sns.pointplot(x="Decision Threshold", y="Values",hue="Metrics", data=plot_df,ax=ax)
```

Out[ ]:

```
<Axes: xlabel='Decision Threshold', ylabel='Values'>
```



Logistic Regression does slightly better than Decision Tree and Random Forest.
Based on the above Test/Train curves, we can keep threshold to 0.4.
Now Finally let's look at Logistic Regression Confusion Matrix

In [ ]:

```
thresh = 0.4 ### Threshold chosen from above Curves
y_pred_test_thresh = logreg_clf.predict_proba(X_test_imp)[:,1]
y_pred = (y_pred_test_thresh > thresh).astype(int)
print("Test Accuracy: ",accuracy_score(y_test,y_pred))
print("Test F1 Score: ",f1_score(y_test,y_pred))
print("Confusion Matrix on Test Data")
pd.crosstab(y_test, y_pred, rownames=['True'], colnames=['Predicted'], margins=True)
```

```
Test Accuracy:  0.8617886178861789
Test F1 Score:  0.9081081081081082
Confusion Matrix on Test Data
```

Out[ ]:

Predicted	0	1	All
True			

False	22	16	38
Predicted	0	1	All
True		84	85
True			
All	23	100	123

Conclusion:

Linear Regression Confusion matrix is very similar to Decision Tree and Random Forest Classifier.In this analysis,we did extensive analysis of input data and were able to achieve Test Accuracy of 86%

Dashboard

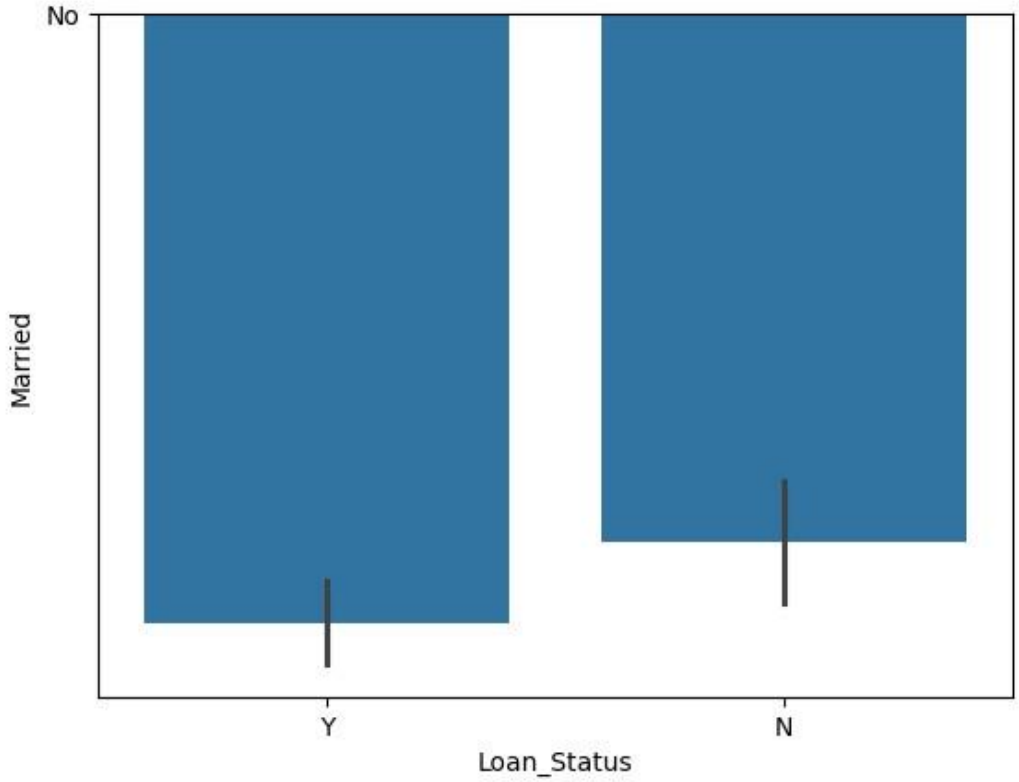
Data visualization

In [ ]:

```
###Barplot###
import seaborn as sns
sns.barplot(x=train_df['Loan_Status'], y=train_df['Married'])
```

Out[ ]:

```
<Axes: xlabel='Loan_Status', ylabel='Married'>
```

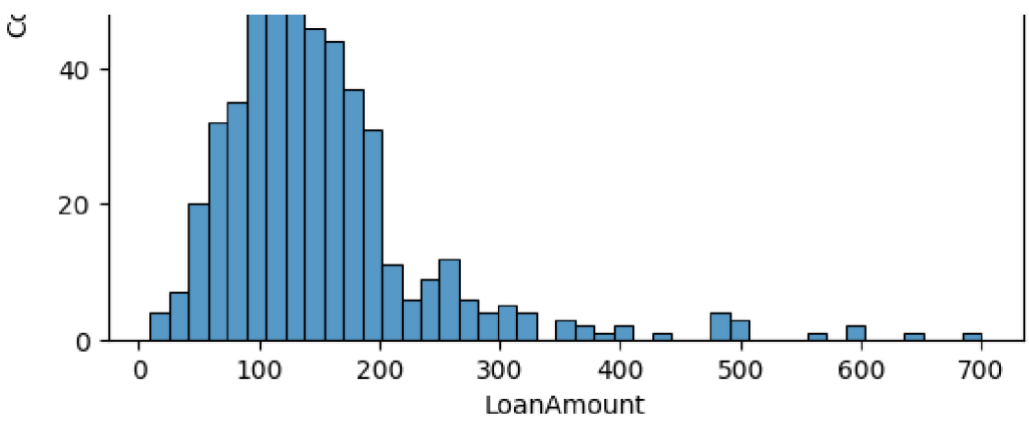


In [ ]:

```
##HistPlot##
sns.histplot(train_df['LoanAmount'])
```

Out[ ]:

```
<Axes: xlabel='LoanAmount', ylabel='Count'>
```

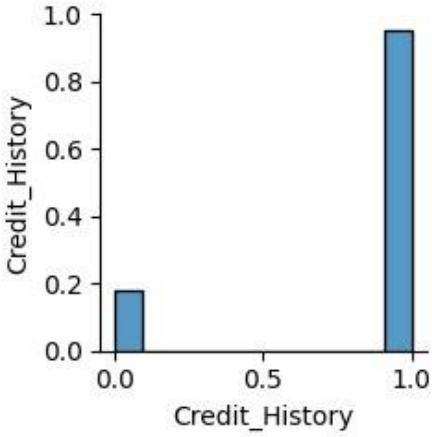


In [ ]:

```
##pairPlot##
sns.pairplot(train_df[['Gender', 'Credit_History', 'Property_Area']])
```

Out[ ]:

```
<seaborn.axisgrid.PairGrid at 0x7b28b30cf280>
```

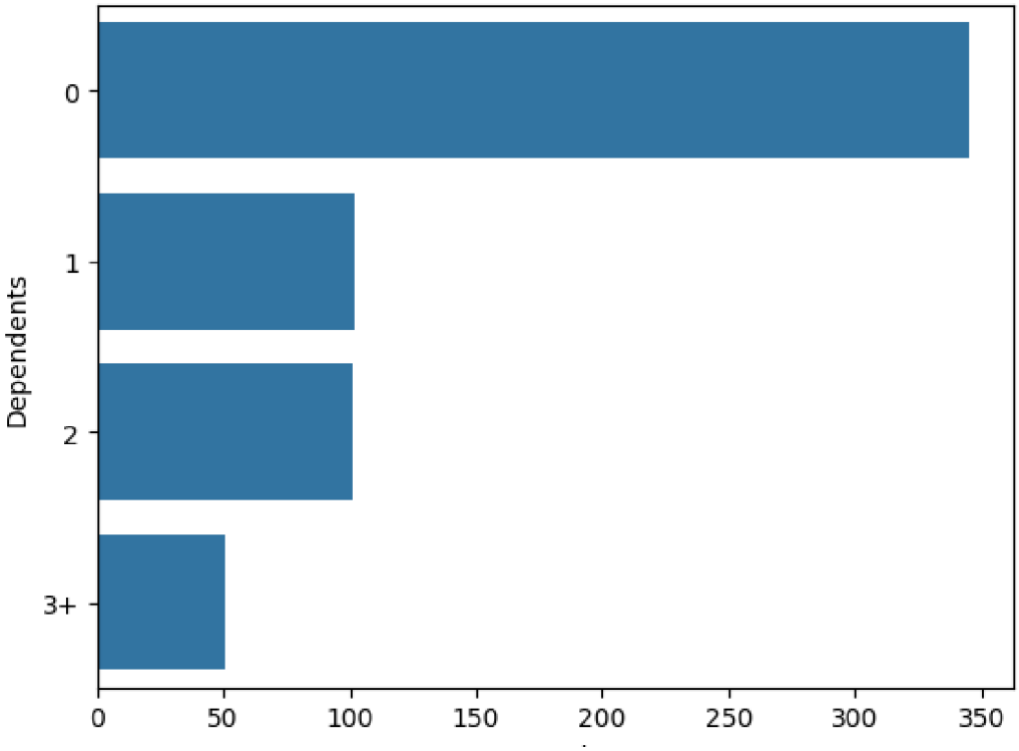


In [ ]:

```
##CountPlot##
sns.countplot(train_df['Dependents'])
```

Out[ ]:

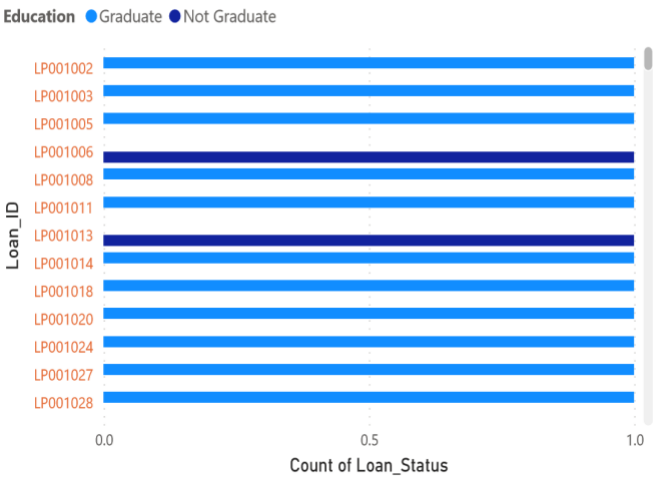
```
<Axes: xlabel='count', ylabel='Dependents'>
```



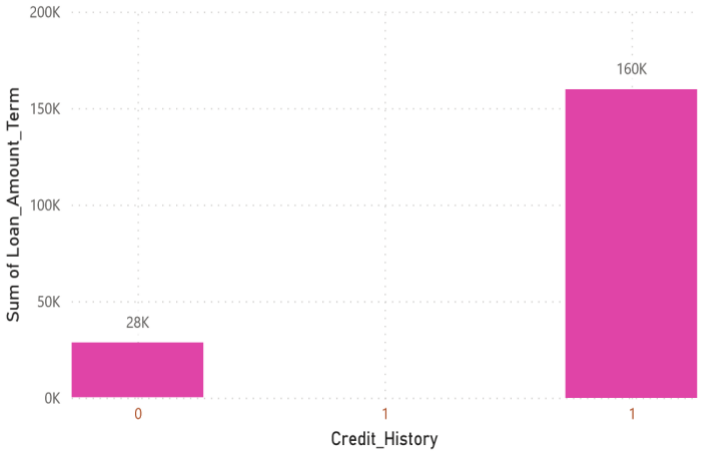
count

Dashboard

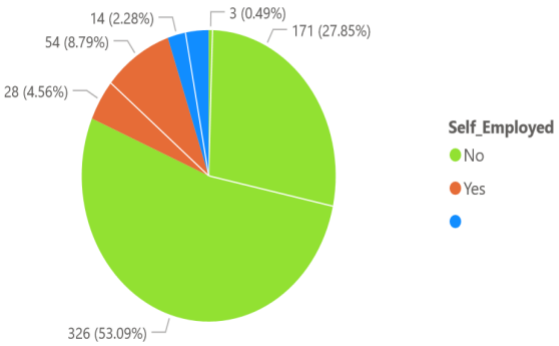
Count of Loan_Status by Loan_ID and Education



Sum of Loan_Amount_Term by Credit_History



Count of Education by Self_Employed and Married



Sum of ApplicantIncome	Sum of CoapplicantIncome	Sum of Credit_History	Sum of Dependents	Education	Gender
23803	0.00	1	3	Graduate	
51763	0.00	1	3	Graduate	
674	5,296.00	1	0	Graduate	
2473	1,843.00	1	0	Graduate	
9833	1,833.00	1	1	Graduate	
16692	0.00	1	0	Graduate	
3583	0.00	1	0	Graduate	
4750	0.00	1	0	Graduate	
9357	0.00	1	3	Graduate	
3750	2,083.00	1	2	Graduate	Female
3667	1,459.00	1	0	Graduate	Female
3410	0.00	1	0	Graduate	Female
4000	2,275.00	1	0	Graduate	Female
3317724	**9,95,444.92**	**475**	**457**		